

ing interval is selected as $s = [n_p / n_{sp}]^{1/2}$. To prevent centering a superpixel on the edge of the image, and to reduce the chances of starting at a noisy point, the initial cluster centers are moved to the lowest gradient position in the 3×3 neighborhood about each center.

The SLIC superpixel algorithm consists of the following steps. Keep in mind that superpixels are vectors in general. When we refer to a “pixel” in the algorithm, we are referring to the (x, y) location of the superpixel relative to the image.

1. **Initialize the algorithm:** Compute the initial superpixel cluster centers,

$$\mathbf{m}_i = [r_i \ g_i \ b_i \ x_i \ y_i]^T, \quad i = 1, 2, \dots, n_{sp}$$

by sampling the image at regular grid steps, s . Move the cluster centers to the lowest gradient position in a 3×3 neighborhood. For each pixel location, p , in the image, set a label $L(p) = -1$ and a distance $d(p) = \infty$.

2. **Assign samples to cluster centers:** For each cluster center $\mathbf{m}_i$, $i = 1, 2, \dots, n_{sp}$, compute the distance, $D_i(p)$ between $\mathbf{m}_i$ and each pixel p in a $2s \times 2s$ neighborhood about $\mathbf{m}_i$. Then, for each p and $i = 1, 2, \dots, n_{sp}$, if $D_i < d(p)$, let $d(p) = D_i$ and $L(p) = i$.
3. **Update the cluster centers:** Let C_i denote the set of pixels in the image with label $L(p) = i$. Update $\mathbf{m}_i$:

$$\mathbf{m}_i = \frac{1}{|C_i|} \sum_{\mathbf{z} \in C_i} \mathbf{z} \quad i = 1, 2, \dots, n_{sp}$$

where $|C_i|$ is the number of pixels in set C_i , and the $\mathbf{z}$'s are given by Eq. (10-86).

4. **Test for convergence:** Compute the Euclidean norms of the differences between the mean vectors in the current and previous steps. Compute the residual error, E , as the sum of the n_{sp} norms. If $E < T$, where T a specified nonnegative threshold, go to Step 5. Else, go back to Step 2.
5. **Post-process the superpixel regions:** Replace all the superpixels in each region, C_i , by their average value, $\mathbf{m}_i$.

Note in Step 5 that superpixels end up as contiguous regions of constant value. The average value is not the only way to compute this constant, but it is the most widely used. For graylevel images, the average is just the average intensity of all the pixels in the region spanned by the superpixel. This algorithm is similar to the k -means algorithm in the previous section, with the exceptions that the distances, D_i , are not specified as Euclidean distances (see below), and that these distances are computed for regions of size $2s \times 2s$, rather than for all the pixels in the image, thus reducing computation time significantly. In practice, SLIC convergence with respect to E can be achieved with fairly large values of T . For example, all results reported by Achanta et al. [2012] were obtained using $T = 10$.

Specifying the Distance Measure

SLIC superpixels correspond to clusters in a space whose coordinates are colors and spatial variables. It would be senseless to use a single Euclidean distance in this case, because the scales in the axes of this coordinate system are different and unrelated. In other words, spatial and color distances must be treated separately. This is accomplished by normalizing the distance of the various components, then combining them into a single measure. Let d_c and d_s denote the color and spatial Euclidean distances between two points in a cluster, respectively:

$$d_c = \left[(r_j - r_i)^2 + (g_j - g_i)^2 + (b_j - b_i)^2 \right]^{1/2} \quad (10-87)$$

and

$$d_s = \left[(x_j - x_i)^2 + (y_j - y_i)^2 \right]^{1/2} \quad (10-88)$$

We then define D as the *composite* distance

$$D = \left[\left(\frac{d_c}{d_{cm}} \right)^2 + \left(\frac{d_s}{d_{sm}} \right)^2 \right]^{1/2} \quad (10-89)$$

where d_{cm} and d_{sm} are the maximum expected values of d_c and d_s . The maximum spatial distance should correspond to the sampling interval; that is, $d_{sm} = s = [n_{tp}/n_{sp}]^{1/2}$. Determining the maximum color distance is not as straightforward, because these distances can vary significantly from cluster to cluster, and from image to image. A solution is to set d_{cm} to a constant c so that Eq. (10-89) becomes

$$D = \left[\left(\frac{d_c}{c} \right)^2 + \left(\frac{d_s}{s} \right)^2 \right]^{1/2} \quad (10-90)$$

We can write this equation as

$$D = \left[d_c^2 + \left(\frac{d_s}{s} \right)^2 c^2 \right]^{1/2} \quad (10-91)$$

This is the distance measure used for each cluster in the algorithm. Constant c can be used to weigh the relative importance between color similarity and spatial proximity. When c is large, spatial proximity is more important, and the resulting superpixels are more compact. When c is small, the resulting superpixels adhere more tightly to image boundaries, but have less regular size and shape.

For grayscale images, as in Example 10.23 below, we use

$$d_c = \left[(I_j - I_i)^2 \right]^{1/2} \quad (10-92)$$